Seulbin **Hwang**

RESEARCHER AT NAVER LABS · AUTONOMOUS DRIVING & ROBOTICS

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Generative world models for autonomous driving and mobile robots

Summary

Researcher at NAVER LABS building world models for autonomous driving and mobile robots, with a particular focus on the realism of dynamic agents. Current work also focuses on real-time world models that remain effective over long horizons, enabling closed-loop training and evaluation of ego planners. Takes a problem-driven approach to research, proactively identifying and solving challenges that deliver tangible value to users of autonomous vehicles and robots. Readily learns and adopts new methods and technologies as the problem demands. Holds master's and bachelor's degrees from KAIST.

Highlights

- **Flow-ERD (IEEE RA-L 2026, accepted), co-first author:** #1 on Waymo's WOSAC 2025 test benchmark (July 2026). Evaluates dynamic-agent realism in world models. Agent-type-aware flow matching and entropy-regularized distillation.
- **Research publications:** IEEE T-ITS (2022; 113 citations): first-author RL for autonomous driving. ICRA (2021; 52 citations): co-authored risk-sensitive RL for mobile robot navigation. OVBEVSeg (ECCV 2026, accepted): 3DGS-based perception for autonomous driving.
- **Research experience:** NAVER LABS, Mar 2020-present. Developed RL navigation for mobile robots serving an entire NAVER office building. Built orderly robot fleet navigation with multi-agent RL. Developed diffusion-based multi-agent simulation for autonomous driving in a government-funded project.

Research Interests

- **Dynamic-agent world modeling:** Realistic dynamic agents. Real-time, long-horizon rollouts for ego-planner training and evaluation.
- **Closed-loop planner training:** Diffusion and RL to bridge the open-loop/closed-loop gap while preserving trajectory diversity.
- **Gaussian splatting for 3D perception:** Open-vocabulary BEV segmentation with 3D geometry (OVBEVSeg). GS and traffic simulation for mobile robot and autonomous driving data augmentation.
- **Reinforcement learning:** Risk-sensitive autonomous navigation. Multi-agent RL for coordinated mobile robot fleets.

Education

Korea Advanced Institute of Science and Technology (KAIST) Mar 2018 - Feb 2020
Master's degree, Graduate School of Mobility

Korea Advanced Institute of Science and Technology (KAIST) Mar 2013 - Feb 2018
Bachelor's degree, Mechanical Engineering

Work Experience

NAVER LABS Corp.
Mar 2020 - Present

Researcher · Full-time

2020-2023 | RL Navigation & Robot Fleet Coordination

- Developed and deployed risk-aware navigation using risk-conditioned distributional RL for mobile robot delivery services throughout a NAVER office building.
- Researched and developed multi-agent RL for orderly robot fleet navigation at delivery pickup points and charging stations.

2024 | Semantic Mapping & Retrieval

- Researched VLM/LLM-based semantic mapping and retrieval to extend delivery robots to a broader range of general-purpose tasks.

2025-2026 | Vectorized Traffic World Models

- Developed vectorized traffic world models for ego-planner training and evaluation in a government-funded autonomous driving project.
- Research aimed to extend NAVER LABS' autonomous shuttle technology operating within a NAVER data center. Developed and published traffic world models capturing realistic, diverse pedestrian, cyclist, and vehicle behavior.

Current | Multi-View Image World Models

- Developing multi-view image-based traffic world models for autonomous vehicles and outdoor robots. Targeting realistic dynamic agents, long-horizon rollouts, and real-time operation.

Selected Publications

Bold denotes my name; * denotes equal contribution.

1. Flow-ERD: Agent-type Aware Flow Matching with Entropy-Regularized Distillation for Diverse Traffic Simulation

Seulbin Hwang*, Kiyoung Om*, Daejung Kim, and Jinhan Lee.

IEEE Robotics and Automation Letters (RA-L), 2026. Accepted. Co-first author. arXiv:2607.06957

2. Open-Vocabulary BEV Segmentation with 3D-Aware Geometric Constraints

Hojun Choi, **Seulbin Hwang**, Daejung Kim, Kisung Kim, Hyunjung Shim, and Jinhan Lee.

European Conference on Computer Vision (ECCV), 2026. Accepted. Paper

3. Autonomous Vehicle Cut-In Algorithm for Lane-Merging Scenarios via Policy-Based Reinforcement Learning Nested Within Finite-State Machine

Seulbin Hwang, Kibeom Lee, Hyeongseok Jeon, and Dongsuk Kum.

IEEE Transactions on Intelligent Transportation Systems, 23(10):17594-17606, 2022. DOI

4. Risk-Conditioned Distributional Soft Actor-Critic for Risk-Sensitive Navigation

Jinyoung Choi, Christopher R. Dance, Jung-eun Kim, **Seulbin Hwang**, and Kyung-sik Park.

IEEE International Conference on Robotics and Automation (ICRA), 2021. DOI